

Sri Lanka Institute of Information Technology



SE3060 - Database Systems Laboratory Worksheet 1

Student ID	Student Name	Email	Contact Number
IT23168404	K.S. Shiwanthi Dias	it23168404@my.sliit.lk	0760369961

1)

- a) Define object types emp_t and dept_t with attributes of EMP and DEPT respectively. Use REF types for workdept (of EMP), and mgrno and admrdept (of DEPT).

```
CREATE TYPE emp_t AS OBJECT(  
    empNo CHAR(6),  
    firstName VARCHAR(12),  
    lastName VARCHAR(15),  
    workDept REF dept_t,  
    sex CHAR(1),  
    birthDate DATE,  
    salary NUMBER(8,2)  
)  
/
```

```
CREATE TYPE dept_t AS OBJECT(  
    deptNo CHAR(3),  
    deptName VARCHAR(36),  
    mgrNo REF emp_t,  
    admrDept REF dept_t  
)  
/
```

- b) Create tables (named as OREMP and ORDEPT) using the object types defined in (a), with appropriate primary keys and referential constraints.

```
CREATE TABLE ORDEPT OF dept_t(  
    deptNo PRIMARY KEY,  
    deptName NOT NULL  
)  
/
```

```
CREATE TABLE OREMP OF emp_t(  
    empNo PRIMARY KEY,  
    firstName NOT NULL,  
    lastName NOT NULL  
)  
/
```

- c) Insert data into the object relational tables created in (b), using the data in EMP and DEPT tables given below. (First insert rows into oremp with null for workdept and rows into ordept with null for admrdept. Then update oremp and ordept with actual REF values for workdept and admrdept.)

```
INSERT INTO OREMP VALUES (  
  emp_t('000010', 'CHRISTINE', 'HAAS', null, 'F', TO_DATE('14/AUG/53', 'DD-MON-  
YYYY'), 72750)  
);
```

```
INSERT INTO OREMP VALUES (  
  emp_t('000020', 'MICHAEL', 'THOMPSON', null, 'M', TO_DATE('02/FEB/68', 'DD-MON-  
YYYY'), 61250)  
);
```

```
INSERT INTO OREMP VALUES (  
  emp_t('000030', 'SALLY', 'KWAN', null, 'F', TO_DATE('11/MAY/71', 'DD-MON-YYYY'),  
58250)  
);
```

```
INSERT INTO OREMP VALUES (  
  emp_t('000060', 'IRVING', 'STERN', null, 'M', TO_DATE('07/JUL/65', 'DD-MON-YYYY'),  
55555)  
);
```

```
INSERT INTO OREMP VALUES (  
  emp_t('000070', 'EVA', 'PULASKI', null, 'F', TO_DATE('26/MAY/73', 'DD-MON-YYYY'),  
56170)  
);
```

```
INSERT INTO OREMP VALUES (  
  emp_t('000050', 'JOHN', 'GEYER', null, 'M', TO_DATE('15/SEP/55', 'DD-MON-YYYY'),  
60175)  
);
```

```
INSERT INTO OREMP VALUES (  
  emp_t('000090', 'EILEEN', 'HENDERSON', null, 'F', TO_DATE('15/MAY/61', 'DD-MON-  
YYYY'), 49750)  
);
```

```
INSERT INTO OREMP VALUES (  
  emp_t('000100', 'THEODORE', 'SPENSER', null, 'M', TO_DATE('18/DEC/76', 'DD-MON-  
YYYY'), 46150)  
);
```

```
INSERT INTO ORDEPT VALUES ('A00', 'SPIFFY COMPUTER SERVICE DIV.',(select  
ref(e) from OREMP e where empNo = '000010' ), NULL);
```

```
INSERT INTO ORDEPT VALUES ('B01', 'PLANNING', (select ref(e) from OREMP e where  
empNo = '000020' ), NULL);
```

```
INSERT INTO ORDEPT VALUES ('C01', 'INFORMATION CENTRE', (select ref(e) from  
OREMP e where empNo = '000030' ), NULL);
```

```
INSERT INTO ORDEPT VALUES ('D01', 'DEVELOPMENT CENTRE', (select ref(e) from  
OREMP e where empNo = '000060' ), NULL);
```

```
UPDATE ORDEPT d  
SET d.ADMRDEPT = (SELECT REF(ad) FROM ORDEPT ad WHERE ad.deptno = 'A00')  
where d.deptno = 'A00';
```

```
UPDATE ORDEPT d  
SET d.ADMRDEPT = (SELECT REF(ad) FROM ORDEPT ad WHERE ad.deptno = 'A00')  
where d.deptno = 'B01';
```

```
UPDATE ORDEPT d  
SET d.ADMRDEPT = (SELECT REF(ad) FROM ORDEPT ad WHERE ad.deptno = 'A00')  
where d.deptno = 'C01';
```

```
UPDATE ORDEPT d  
SET d.ADMRDEPT = (SELECT REF(ad) FROM ORDEPT ad WHERE ad.deptno = 'C01')  
where d.deptno = 'D01';
```

```
UPDATE OREMP e  
SET e.workDept = (SELECT REF(ad) FROM ORDEPT ad WHERE ad.deptno = 'A00')  
where e.empno = '000010';
```

```
UPDATE OREMP e  
SET e.workDept = (SELECT REF(ad) FROM ORDEPT ad WHERE ad.deptno = 'B01')  
where e.empno = '000020';
```

```
UPDATE OREMP e  
SET e.workDept = (SELECT REF(ad) FROM ORDEPT ad WHERE ad.deptno = 'C01')  
where e.empno = '000030';
```

```
UPDATE OREMP e  
SET e.workDept = (SELECT REF(ad) FROM ORDEPT ad WHERE ad.deptno = 'D01')  
where e.empno = '000060';
```

```
UPDATE OREMP e  
SET e.workDept = (SELECT REF(ad) FROM ORDEPT ad WHERE ad.deptno = 'D01')  
where e.empno = '000070';
```

```
UPDATE OREMP e
SET e.workDept = (SELECT REF(ad) FROM ORDEPT ad WHERE ad.deptno = 'C01')
where e.empno = '000050';
```

```
UPDATE OREMP e
SET e.workDept = (SELECT REF(ad) FROM ORDEPT ad WHERE ad.deptno = 'B01')
where e.empno = '000090';
```

```
UPDATE OREMP e
SET e.workDept = (SELECT REF(ad) FROM ORDEPT ad WHERE ad.deptno = 'B01')
where e.empno = '000100';
```

2)

- a) Get the department name and manager's lastname for all departments.

```
SELECT d.DEPTNAME , d.MGRNO.LASTNAME
FROM ordept d;
```

- b) Get the employee number, lastname and the department name of every employee

```
SELECT e.EMPNO , e.LASTNAME, e.WORKDEPT.DEPTNAME
FROM oremp e
```

- c) For each department, display the department number, department name, and name of the administrative department.

```
SELECT d.DEPTNO, d.DEPTNAME , d.ADMRDEPT.DEPTNAME
FROM ordept d;
```

- d) For each department, display the department number, department name, the name of the administrative department and the last name of the manager of the administrative department

```
SELECT d.DEPTNO, d.DEPTNAME , d.ADMRDEPT.DEPTNAME,
d.ADMRDEPT.MGRNO.LASTNAME
FROM ordept d;
```

- e) Display the employee number, firstname, lastname and salary of every employee, along with lastname and salary of the manager of the employee's work department.

```
SELECT      e.EMPNO      ,e.FIRSTNAME,      e.LASTNAME,      e.SALARY,
e.WORKDEPT.MGRNO.LASTNAME,e.WORKDEPT.MGRNO.SALARY
FROM oremp e
```

- f) Show the average salary for men and the average salary for women for each department. Identify the department by both department number and name.

```
SELECT
e.workdept.deptNo  AS dept_number,
e.workdept.deptName AS dept_name,
ROUND(SUM(CASE WHEN e.sex = 'F' THEN e.salary END) /
COUNT(CASE WHEN e.sex = 'F' THEN 1 END), 2) AS avg_female_salary,
ROUND(SUM(CASE WHEN e.sex = 'M' THEN e.salary END) /
COUNT(CASE WHEN e.sex = 'M' THEN 1 END), 2) AS avg_male_salary
FROM
  oremp e
GROUP BY
e.workdept.deptNo,
e.workdept.deptName
ORDER BY
e.workdept.deptNo;
```

3)

b)Execute the SQL script as follows and study the result.

SQL>@Z:\script

```

Command Prompt - docker e x + v

DEPTNAME                                MGRNO.LASTNAME
-----
SPIFFY COMPUTER SERVICE DIV.           HAAS
PLANNING                               THOMPSON
INFORMATION CENTRE                     KWAN
DEVELOPMENT CENTRE                     STERN

EMPNO  LASTNAME                                WORKDEPT.DEPTNAME
-----
000010 HAAS                                SPIFFY COMPUTER SERVICE DIV.
000020 THOMPSON                            PLANNING
000030 KWAN                                INFORMATION CENTRE
000060 STERN                              DEVELOPMENT CENTRE
000070 PULASKI                            DEVELOPMENT CENTRE
000050 GEYER                              INFORMATION CENTRE
000090 HENDERSON                          PLANNING
000100 SPENSER                            PLANNING

8 rows selected.

DEP DEPTNAME                                ADMRDEPT.DEPTNAME
-----
A00 SPIFFY COMPUTER SERVICE DIV.           SPIFFY COMPUTER SERVICE DIV.
B01 PLANNING                               SPIFFY COMPUTER SERVICE DIV.
C01 INFORMATION CENTRE                     SPIFFY COMPUTER SERVICE DIV.
D01 DEVELOPMENT CENTRE                     INFORMATION CENTRE

DEP DEPTNAME                                ADMRDEPT.DEPTNAME
-----
ADMIRDEPT.MGRNO.
A00 SPIFFY COMPUTER SERVICE DIV.           SPIFFY COMPUTER SERVICE DIV.
HAAS
B01 PLANNING                               SPIFFY COMPUTER SERVICE DIV.
HAAS

```

```

Command Prompt - docker e x + v

DEP DEPTNAME                                ADMRDEPT.DEPTNAME
-----
ADMIRDEPT.MGRNO.
A00 SPIFFY COMPUTER SERVICE DIV.           SPIFFY COMPUTER SERVICE DIV.
HAAS
B01 PLANNING                               SPIFFY COMPUTER SERVICE DIV.
HAAS
C01 INFORMATION CENTRE                     SPIFFY COMPUTER SERVICE DIV.
HAAS

DEP DEPTNAME                                ADMRDEPT.DEPTNAME
-----
ADMIRDEPT.MGRNO.
D01 DEVELOPMENT CENTRE                     INFORMATION CENTRE
KWAN

EMPNO  FIRSTNAME  LASTNAME                                SALARY WORKDEPT.MGRNO.
-----
WORKDEPT.MGRNO.SALARY
000010 CHRISTINE  HAAS                                72750 HAAS
72750
000020 MICHAEL    THOMPSON                            61250 THOMPSON
61250
000030 SALLY         KWAN                                58250 KWAN
58250

EMPNO  FIRSTNAME  LASTNAME                                SALARY WORKDEPT.MGRNO.
-----
WORKDEPT.MGRNO.SALARY

```

```
Command Prompt - docker e x + v
58250

EMPNO FIRSTNAME LASTNAME SALARY WORKDEPT.MGRNO.
-----
WORKDEPT.MGRNO.SALARY
000060 IRVING STERN 55555 STERN
000070 EVA PULASKI 56170 STERN
000050 JOHN GEYER 60175 KWAN
58250

EMPNO FIRSTNAME LASTNAME SALARY WORKDEPT.MGRNO.
-----
WORKDEPT.MGRNO.SALARY
000090 EILEEN HENDERSON 49750 THOMPSON
000100 THEODORE SPENSER 46150 THOMPSON
61250

8 rows selected.

DEP DEPT_NAME AVG_FEMALE_SALARY AVG_MALE_SALARY
---
A00 SPIFFY COMPUTER SERVICE DIV. 72750
B01 PLANNING 49750 53700
C01 INFORMATION CENTRE 58250 60175
D01 DEVELOPMENT CENTRE 56170 55555

SQL> set echo on
SQL> @/tmp/scriptor.sql
SQL>
```

c)Set echo on and execute the script again and compare the result with the result you got in the previous step. Try to understand what would happen when you set echo on and off

```
SQL>set echo on
SQL>@Z:\script
```

```
Command Prompt - docker e x + v

SQL>
SQL>
SQL> SELECT d.DEPTNAME , d.MGRNO.LASTNAME
2 FROM ordept d
3 /

DEPTNAME MGRNO.LASTNAME
-----
SPIFFY COMPUTER SERVICE DIV. HAAS
PLANNING THOMPSON
INFORMATION CENTRE KWAN
DEVELOPMENT CENTRE STERN

SQL>
SQL>
SQL> SELECT e.EMPNO , e.LASTNAME, e.WORKDEPT.DEPTNAME
2 FROM oremp e
3 /

EMPNO LASTNAME WORKDEPT.DEPTNAME
-----
000010 HAAS SPIFFY COMPUTER SERVICE DIV.
000020 THOMPSON PLANNING
000030 KWAN INFORMATION CENTRE
000060 STERN DEVELOPMENT CENTRE
000070 PULASKI DEVELOPMENT CENTRE
000050 GEYER INFORMATION CENTRE
000090 HENDERSON PLANNING
000100 SPENSER PLANNING

8 rows selected.

SQL>
SQL>
SQL>
SQL> SELECT d.DEPTNO, d.DEPTNAME , d.ADMRDEPT.DEPTNAME
2 FROM ordept d
3 /
```



```
Command Prompt - docker x + v
SQL>
SQL>
SQL> SELECT d.DEPTNO, d.DEPTNAME, d.ADMRDEPT.DEPTNAME
2 FROM ordept d
3 /

DEP DEPTNAME                                ADMRDEPT.DEPTNAME
-----
A00 SPIFFY COMPUTER SERVICE DIV.            SPIFFY COMPUTER SERVICE DIV.
B01 PLANNING                                SPIFFY COMPUTER SERVICE DIV.
C01 INFORMATION CENTRE                      SPIFFY COMPUTER SERVICE DIV.
D01 DEVELOPMENT CENTRE                      INFORMATION CENTRE

SQL>
SQL>
SQL>
SQL> SELECT d.DEPTNO, d.DEPTNAME, d.ADMRDEPT.DEPTNAME, d.ADMRDEPT.MGRNO, LASTNAME
2 FROM ordept d
3 /

DEP DEPTNAME                                ADMRDEPT.DEPTNAME
-----
ADMNDEPT.MGRNO.
-----
A00 SPIFFY COMPUTER SERVICE DIV.            SPIFFY COMPUTER SERVICE DIV.
HAAS
B01 PLANNING                                SPIFFY COMPUTER SERVICE DIV.
HAAS
C01 INFORMATION CENTRE                      SPIFFY COMPUTER SERVICE DIV.
HAAS

DEP DEPTNAME                                ADMRDEPT.DEPTNAME
-----
ADMNDEPT.MGRNO.
-----
D01 DEVELOPMENT CENTRE                      INFORMATION CENTRE
KWAN
```

- d) Set termout off and execute the script again and compare the result with the results you got in the previous steps. Try to understand what would happen when you set termout on and off

SQL>set termout on SQL>@Z:\script

This command prevents displaying output on the screen, but the script still runs silently. It is useful for long scripts or background tasks

- e) Set termout off and echo on and run the script on Oracle with spooling. Then open the spooled file in a text editor and study it.

```
File Edit View
|

SQL>
SQL>
SQL> SELECT d.DEPTNAME , d.MGRNO.LASTNAME
2 FROM ordept d
3 /

DEPTNAME                MGRNO.LASTNAME
-----
SPIFFY COMPUTER SERVICE DIV.  HAAS
PLANNING                   THOMPSON
INFORMATION CENTRE          KWAN
DEVELOPMENT CENTRE          STERN

SQL>
SQL>
SQL> SELECT e.EMPNO , e.LASTNAME, e.WORKDEPT.DEPTNAME
2 FROM orepmp e
3 /

EMPNO  LASTNAME      WORKDEPT.DEPTNAME
-----
000010 HAAS          SPIFFY COMPUTER SERVICE DIV.
000020 THOMPSON    PLANNING
000030 KWAN         INFORMATION CENTRE
000060 STERN          DEVELOPMENT CENTRE
000070 PULASKI       DEVELOPMENT CENTRE
000050 GEYER          INFORMATION CENTRE
000090 HENDERSON      PLANNING
000100 SPENSER       PLANNING

8 rows selected.

SQL>
SQL>
SQL>
SQL>
SQL> SELECT d.DEPTNO, d.DEPTNAME , d.ADMRDEPT.DEPTNAME
2 FROM ordept d

Ln 1, Col 1 7,991 characters Plain text 100% Unix (LF) UTF-8
```

4)

```
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL>
SQL> drop type emp_t force
2 /

Type dropped.

SQL> drop table OREMP cascade constraint
2 /

Table dropped.

SQL>
```